

CHAMPAIGN WILLARD, IL, USA

WMO: 725315

Lat: 40.040N

Lon: 88.278W

Elev: 230

StdP: 98.59

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94870

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.7	-15.7	-22.4	0.5	-18.1	-19.4	0.7	-14.9	14.2	-1.8	12.7	-2.2	4.4	260	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	33.0	23.9	32.0	23.7	30.6	23.0	26.2	30.9	25.2	29.8	24.3	28.4	4.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.8	20.4	29.3	23.8	19.2	28.1	22.9	18.1	26.9	82.4	31.0	78.1	29.7	74.4	28.5	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	11.0	9.6	-23.0	35.0	3.8	1.1	-25.7	35.8	-27.9	36.5	-30.0	37.1	-32.7	37.9		
(5)			-23.1	27.8	3.6	0.9	-25.7	28.4	-27.8	29.0	-29.8	29.4	-32.3	30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	-3.6	-1.1	4.8	11.5	17.6	22.4	23.5	22.5	19.5	12.7	5.4	-0.6		
	DBStd	10.86	6.91	6.48	6.19	5.28	4.87	3.50	2.95	2.78	4.15	5.33	5.58	6.06		
	HDD10.0	1489	422	312	186	44	3	0	0	0	1	32	158	332		
	HDD18.3	3142	678	543	421	212	76	8	1	3	35	189	388	587		
	CDD10.0	1954	1	3	24	89	238	371	418	386	286	116	21	3		
	CDD18.3	566	0	0	1	7	52	129	161	131	71	14	1	0		
	CDH23.3	5214	0	0	12	93	516	1191	1474	1096	686	144	2	0		
	CDH26.7	1796	0	0	1	17	172	439	531	348	251	37	0	0		
(14) Wind	WSAvg	4.6	5.6	5.5	5.5	5.8	5.1	4.0	3.1	2.8	3.2	4.4	5.2	5.2		
(15) Precipitation	PrecAvg	1003	52	47	80	97	111	101	113	105	84	70	81	66		
	PrecMax	1487	158	115	197	243	257	232	351	254	221	184	256	168		
	PrecMin	683	2	8	19	15	15	8	32	32	8	4	14	7		
	PrecStd	147	36	25	42	53	58	56	70	58	52	40	47	41		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.8	18.0	25.1	28.7	32.8	34.1	35.0	33.5	33.9	30.2	22.5	17.3		
		MCWB	13.7	13.3	17.2	17.9	21.3	23.3	26.0	24.9	22.6	20.9	15.0	15.0		
	2%	DB	12.2	14.2	21.1	26.2	30.6	32.6	32.7	31.9	31.9	27.3	18.9	13.7		
		MCWB	11.0	11.5	15.1	16.7	20.4	23.2	25.5	24.5	21.9	18.6	14.4	11.7		
	5%	DB	8.9	11.2	17.6	23.4	28.7	31.2	31.3	30.1	29.6	24.3	16.8	10.7		
		MCWB	7.3	8.8	13.0	15.8	19.5	22.9	24.9	23.7	21.1	17.1	13.4	9.1		
	10%	DB	5.5	7.8	14.6	20.9	26.4	29.3	29.7	28.6	27.5	21.7	14.0	7.7		
		MCWB	4.1	5.7	11.1	14.6	18.7	21.9	23.7	22.8	19.7	16.3	10.7	6.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.0	15.1	18.0	20.2	23.6	26.2	27.8	26.9	24.9	22.2	17.5	15.6		
		MCDB	15.3	17.2	23.7	26.1	29.2	31.1	32.5	31.4	30.7	28.1	19.8	17.2		
	2%	WB	11.1	12.0	16.0	18.4	22.2	25.0	26.6	25.6	23.6	20.4	15.5	12.1		
		MCDB	11.9	13.8	19.8	22.9	28.0	30.2	31.4	30.2	28.9	24.3	18.1	13.5		
	5%	WB	7.6	8.8	14.0	17.0	21.0	24.0	25.7	24.6	22.4	18.8	13.8	9.0		
		MCDB	8.7	11.0	17.1	21.4	26.1	28.9	30.1	28.7	27.0	22.6	16.3	10.2		
	10%	WB	4.0	5.9	11.5	15.4	19.9	23.1	24.7	23.7	21.3	17.0	11.4	6.3		
		MCDB	5.3	7.9	14.3	19.9	24.6	27.8	28.6	27.3	25.4	20.8	13.8	7.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.0	10.6	12.3	11.8	11.1	10.7	11.0	12.9	12.2	10.1	7.9		
		MCDBR	11.3	12.9	14.5	15.7	14.4	12.7	11.7	12.5	14.9	15.3	13.3	10.8		
	5% WB	MCWBR	9.8	10.3	9.2	8.5	6.3	5.6	5.9	6.3	6.4	7.7	9.2	9.2		
		MCWBR	11.0	12.3	13.1	13.2	12.2	10.6	10.5	10.7	12.2	12.6	12.0	10.0		
(40) Clear-Sky Solar Irradiance	taub		0.301	0.317	0.353	0.395	0.435	0.439	0.459	0.457	0.414	0.361	0.314	0.298		
	taud		2.436	2.384	2.365	2.253	2.195	2.225	2.152	2.147	2.245	2.393	2.483	2.460		
	Ebn at Noon		866	900	899	879	847	841	820	810	822	836	843	840		
	Edh at Noon		80	97	111	133	144	140	149	146	124	96	76	73		
(44) All-Sky Solar Radiation	RadAvg		1.79	2.66	3.65	4.76	5.43	6.24	6.15	5.48	4.62	3.18	2.12	1.51		
	RadStd		0.20	0.27	0.29	0.34	0.46	0.40	0.34	0.34	0.34	0.34	0.20	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page